

CampusGuard

Members:

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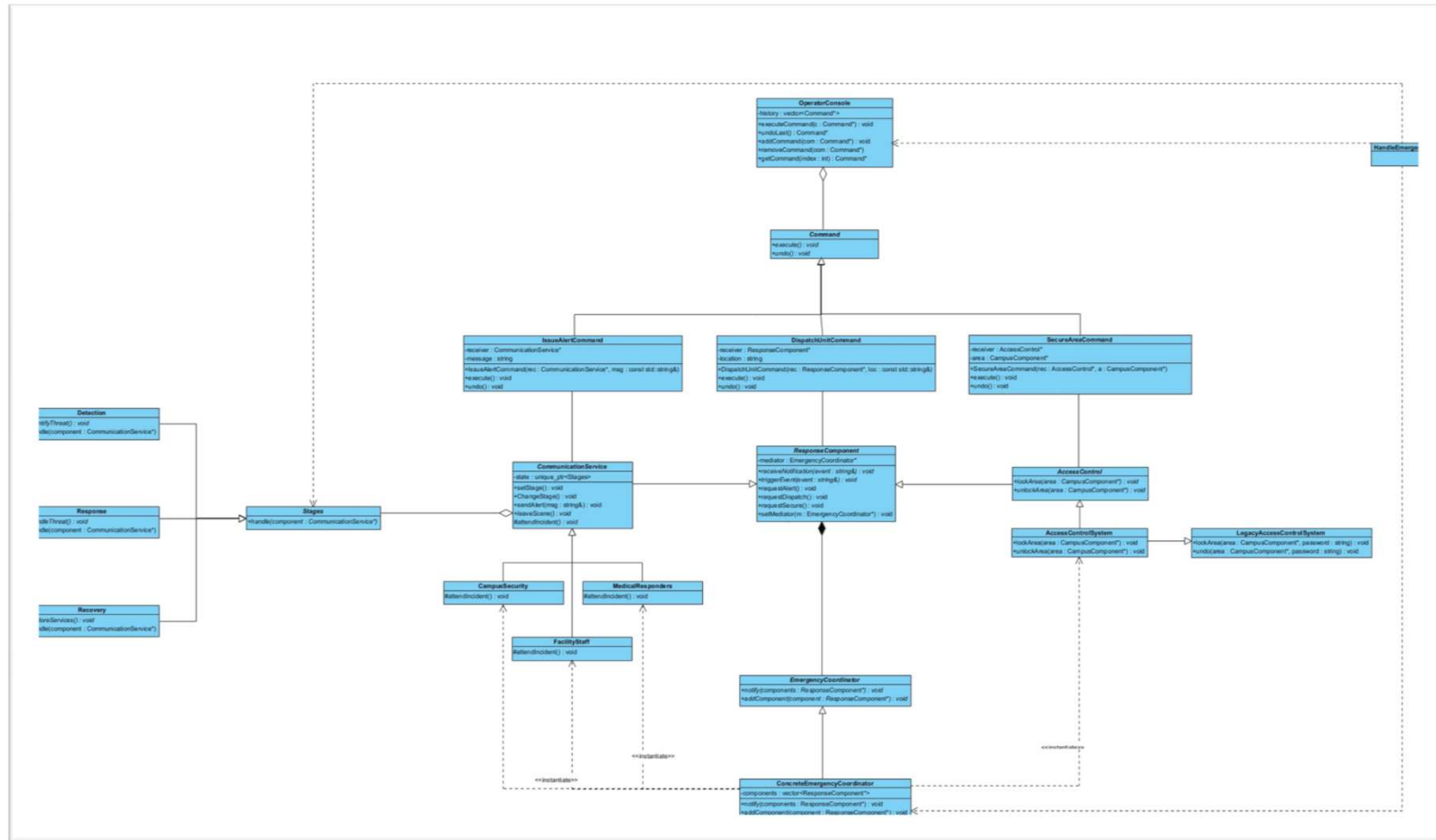
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System Description

CampusGuard is an emergency-response coordination platform designed for a large university campus. The system facilitates the efficient and decoupled interaction between various emergency units—including campus security, medical responders, and access control systems—through a central mediator. It manages the lifecycle of incidents via state transitions and standardizes emergency response workflows using predefined template methods and executable commands, ultimately allowing operators to seamlessly trigger, coordinate, and undo critical emergency actions.

UML



GoF Participant Mapping

Command

- **Command:** Command (Interface)
- **ConcreteCommand:** DispatchUnitCommand, SecureAreaCommand, IssueAlertCommand
- **Invoker:** OperatorConsole
- **Receiver:** Various ResponseComponent subclasses (e.g., CommunicationService, AccessControlSystem)

Mediator

- **Mediator:** EmergencyCoordinator
- **ConcreteMediator:** ConcreteEmergencyCoordinator
- **Colleague:** ResponseComponent
- **ConcreteColleague:** CommunicationService, AccessControl, CampusSecurity, MedicalResponders

GoF Participant Mapping

Adapter

- **Target:** AccessControl
- **Adapter:** AccessControlSystem
- **Adaptee:** LegacyAccessControlSystem (External/Legacy Service)

Facade

- **Facade:** HandleEmergency (Acts as the high-level entry point orchestrating subsystems)
- **Subsystems:** OperatorConsole, ConcreteEmergencyCoordinator, and Stages (State context).

GoF Participant Mapping

State

- **Context:** CommunicationService
- **State:** Stages
- **ConcreteState:** Detection, Response, Recovery

Template Method

- **AbstractClass:** CommunicationService
- **Template Method:** triggerEvent()
- **PrimitiveOperations:** attendIncident(), leaveScene()
- **ConcreteClass:** CampusSecurity, MedicalResponders, FacilityStaff

Justification for the Two Team-Selected Patterns

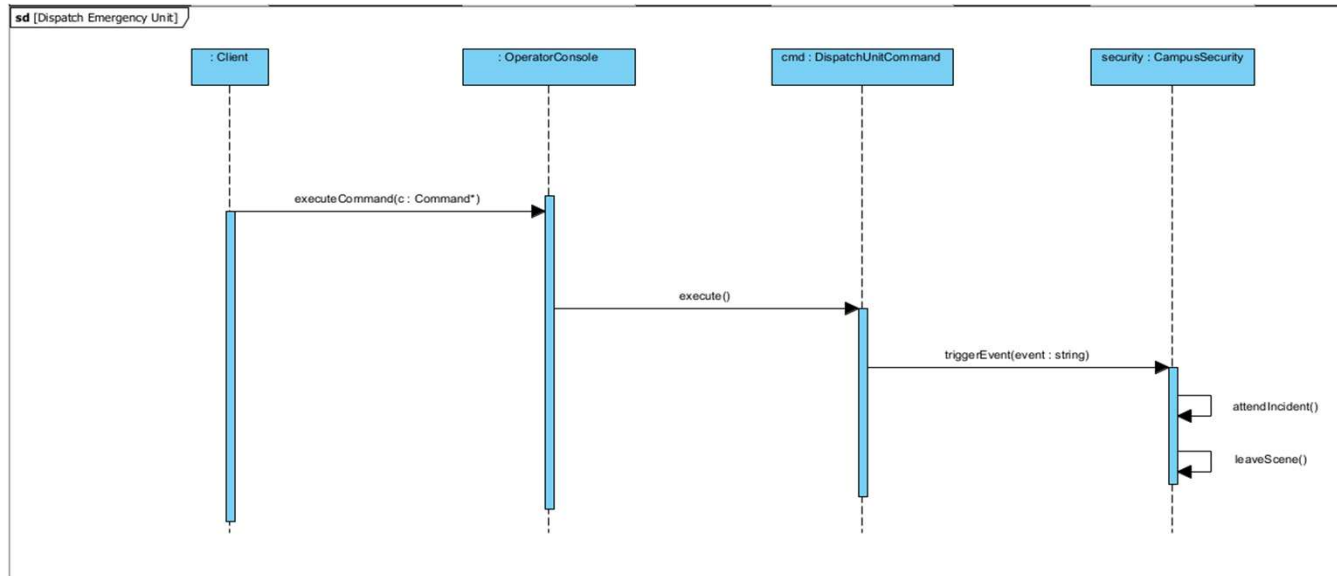
State Pattern

- **Problem Solved:** The scenario requires registering incidents and dynamically changing their operational status over time. **Why it's better:** Without State, the CommunicationService would require massive conditional statements (e.g., switch or if/else) to check if the incident is in detection, response, or recovery mode before allowing specific actions. The State pattern encapsulates state-specific behavior, allowing the component's behavior to change cleanly at runtime based on the incident's current operational phase.

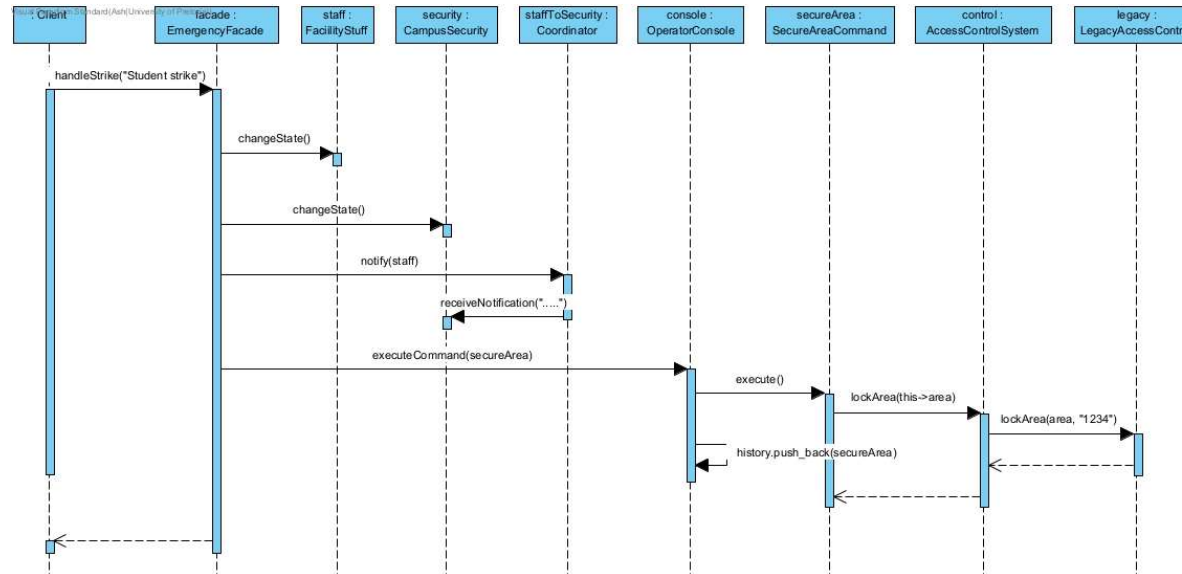
Template Method

- **Problem Solved:** Different emergency response teams (e.g., Security, Medical, Facilities) have a shared overarching workflow for handling incidents, but the specific implementation details of assessing and attending the scene differ. **Why it's better:** Instead of having each response unit duplicate the workflow logic, the Template Method enforces a strict sequence of operations (attendIncident(), leaveScene()) in the base CommunicationService via the triggerEvent() method. Subclasses only override the specific steps, reducing code duplication and enforcing a standardized protocol.

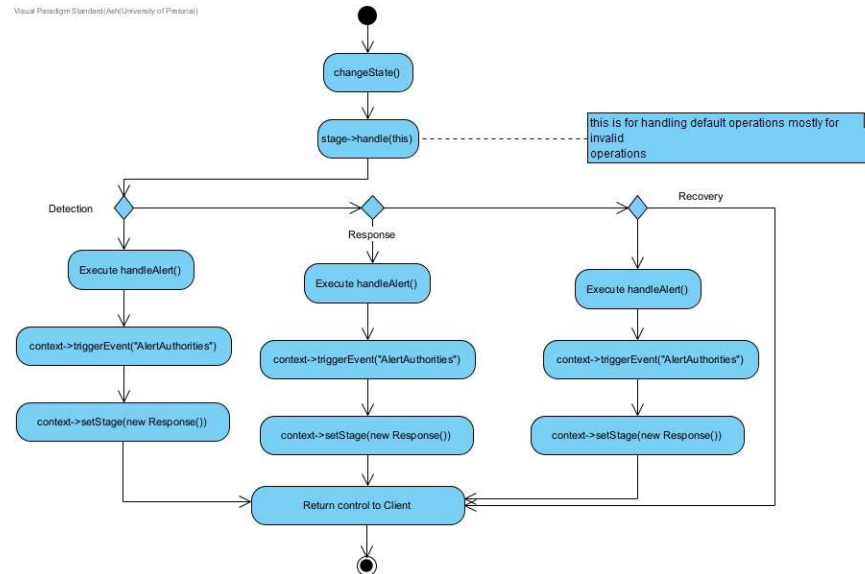
Behavioural Diagrams – Dispatch Emergency Unit



Behavioural Diagrams - Facade



Behavioural Diagrams - Activity diagram



Design and Ownership Rationale

- **Decoupling via Mediator:** Direct dependencies between response units are eliminated by forcing all cross-component communication through the ConcreteEmergencyCoordinator.
- **Command Encapsulation:** Actions are decoupled from execution. The OperatorConsole dynamically manages a history of commands, making undo() operations trivial without needing to know the specifics of the ResponseComponent.
- **Memory Management & Ownership:** All polymorphic base classes define virtual destructors to prevent resource leaks during deletion. The Mediator maintains non-owning pointers to Colleagues, while the OperatorConsole owns the Command objects dynamically allocated to its history list. std::unique_ptr is utilized to strictly own the Stages (State pattern) within the CommunicationService, ensuring clean and deterministic memory cleanup.

GDB Investigation

```
+++++
CommunicationService received: Student strike outside the school
CommunicationService triggered: Student strike outside the school
+++++
+ Stage: Incident Response +
+ Report_Received: Suspicious activity, Lot C +
+ Officer_Dispatched: Officer M. Dlamini +
+ Action_Taken: Approached, questioned subject +
+ Outcome: Verified student ID, no incident +
+++++
+ Stage: Scene Assessment +
+ Hazards_Checked: Fire, obstruction, crowd risk +
+ Threat_Level: Low +
+ Access_Points: 2 identified +
+ Decision: Safe to proceed, no backup +
+++++

AccessControlSystem is locking the area.
This area is locked from access, please try again later.

Thread 1 received signal SIGSEGV, Segmentation fault.
0x00007ff7f2e63f27 in ResponseComponent::~ResponseComponent (this=0x69a970)
    at C:/Users/matmo/OneDrive/Documents/COS214_Practical5/ResponseComponent.h:17
17      delete mediator;
(gdb) |
```

```
+ Hazards_Checked: Traffic, fire, crowd control +
+ Patients_Identified: 1 +
+ Scene_Safety: Confirmed Safe +
+ Decision: Proceed with treatment +
+++++

CommunicationService received: Student strike outside the school
CommunicationService triggered: Student strike outside the school
+++++
+ Stage: Incident Response +
+ Report_Received: Suspicious activity, Lot C +
+ Officer_Dispatched: Officer M. Dlamini +
+ Action_Taken: Approached, questioned subject +
+ Outcome: Verified student ID, no incident +
+++++

+ Stage: Scene Assessment +
+ Hazards_Checked: Fire, obstruction, crowd risk +
+ Threat_Level: Low +
+ Access_Points: 2 identified +
+ Decision: Safe to proceed, no backup +
+++++

AccessControlSystem is locking the area.
This area is locked from access, please try again later.
[Thread 22420.0x583c exited with code 0]
[Thread 22420.0x42e8 exited with code 0]
[Thread 22420.0x403c exited with code 0]
[Inferior 1 (process 22420) exited normally]
(gdb) |
```

GDB Investigation

```
15 public:
16     virtual ~ResponseComponent() {
17         delete mediator;
18         state = nullptr;
19     }
20
```

```
Thread 1 hit breakpoint 2, 0x0000773ff20c6 in main () at main.cpp:72
72 int main() {
(gdb) step
75     std::string issue1 = "Student strike outside the school";
(gdb) step
std::allocator<char>::allocator (this=optimized out) at C:/msys64/ucrt64/include/c++/16.2.0/bits/allocator.h:168
168     allocator() _GLIBCXX_NOTHROW { }
(gdb) step
std::_new_allocator<char>::_new_allocator (this=0x5ffe3e) at C:/msys64/ucrt64/include/c++/16.2.0/bits/new_allocator.h:88
88     _new_allocator() _GLIBCXX_USE_NOEXCEPT { }
(gdb) step
std::allocator<char>::allocator (this=optimized out) at C:/msys64/ucrt64/include/c++/16.2.0/bits/allocator.h:168
168     allocator() _GLIBCXX_NOTHROW { }
(gdb) step
main () at main.cpp:75
75     std::string issue1 = "Student strike outside the school";
(gdb) step
std::_cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >::basic_string (this=0x5ffe10,
_0x0000773ff20c6 <_gnu_cxx::default_lock_policy=66> "Student strike outside the school", ...)
at C:/msys64/ucrt64/include/c++/16.2.0/bits/basic_string.h:731
731     : _M_dataplus(_M_local_data(), ...)
(gdb) step
std::_cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >::_M_local_data (this=0x5ffe10)
at C:/msys64/ucrt64/include/c++/16.2.0/bits/basic_string.h:245
245     return std::pointer_traits<pointer>::pointer_to(*_M_local_buf);
(gdb) next
249     }
(gdb) next
std::_cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >::basic_string (this=0x5ffe10,
_0x0000773ff20c6 <_gnu_cxx::default_lock_policy=66> "Student strike outside the school", ...)
at C:/msys64/ucrt64/include/c++/16.2.0/bits/basic_string.h:734
734     if (_S == 0)
(gdb) next
737     const_char_t* __end = _S + traits_type::length(_S);
(gdb) backtrace
#0 std::_cxx11::basic_string<char, std::char_traits<char>, std::allocator<char> >::basic_string (this=0x5ffe10,
_0x0000773ff20c6 <_gnu_cxx::default_lock_policy=66> "Student strike outside the school", ...)
at C:/msys64/ucrt64/include/c++/16.2.0/bits/basic_string.h:737
#1 0x0000773ff20c6 in main () at main.cpp:75
(gdb) print issue1
No symbol "issue1" in current context.
(gdb) print issue1
No symbol "issue1" in current context.
(gdb) next
738     _M_construct(_S, __end, forward_iterator_tag());
(gdb) next
739     }
(gdb) print issue1
No symbol "issue1" in current context.
(gdb) next
```

Valgrind Evidence

Before

vs

After

```

+-----+
+ Stage:                               Scene Assessment +
+ Hazards_Checked:                     Fire, obstruction, crowd risk +
+ Threat_Level:                        Low +
+ Access_Points:                       2 identified +
+ Decision:                           Safe to proceed, no backup +
+-----+

AccessControlSystem is locking the area.
This area is locked from access, please try again later.
==2003==
==2003== HEAP SUMMARY:
==2003==   in use at exit: 120 bytes in 4 blocks
==2003==   total heap usage: 35 allocs, 31 frees, 80,282 bytes allocated
==2003==
==2003== 120 (24 direct, 96 indirect) bytes in 1 blocks are definitely lost in loss record 4 of 4
==2003==   at 0x4846FA3: operator new(unsigned long) (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==2003==   by 0x10B862: main (main.cpp:87)
==2003==
==2003== LEAK SUMMARY:
==2003==   definitely lost: 24 bytes in 1 blocks
==2003==   indirectly lost: 96 bytes in 3 blocks
==2003==   possibly lost: 0 bytes in 0 blocks
==2003==   still reachable: 0 bytes in 0 blocks
==2003==   suppressed: 0 bytes in 0 blocks
==2003==
==2003== For lists of detected and suppressed errors, rerun with: -s
==2003== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
mat@ONE2ONECAPITAL:~/COS214 Practical5$
```

```

CommunicationService received: Student strike outside the school
CommunicationService triggered: Student strike outside the school
+-----+
+ Stage:                               Incident Response +
+ Report_Received:                     Suspicious activity, Lot C +
+ Officer_Dispatched:                  Officer M. Dlamini +
+ Action_Taken:                        Approached, questioned subject +
+ Outcome:                             Verified student ID, no incident +
+-----+

+ Stage:                               Scene Assessment +
+ Hazards_Checked:                     Fire, obstruction, crowd risk +
+ Threat_Level:                        Low +
+ Access_Points:                       2 identified +
+ Decision:                           Safe to proceed, no backup +
+-----+

AccessControlSystem is locking the area.
This area is locked from access, please try again later.
==2617==
==2617== HEAP SUMMARY:
==2617==   in use at exit: 0 bytes in 0 blocks
==2617==   total heap usage: 34 allocs, 34 frees, 80,218 bytes allocated
==2617==
==2617== All heap blocks were freed -- no leaks are possible
==2617==
==2617== For lists of detected and suppressed errors, rerun with: -s
==2617== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
mat@ONE2ONECAPITAL:~/COS214 Practical5$
```

Docker and Execution Instructions

Dockerfile

```
FROM gcc:11
WORKDIR /app
COPY . .
RUN make clean && make
CMD ["/CampusGuard"]
```

docker-compose.yml

```
version: '3'
services:
  campusguard:
    build: .
    image: campusguard_app
```

To build and run the complete application inside Docker for the demonstration, run the following command in the root directory:

```
docker compose up --build
```

GitHub Repository Link and Collaboration Reflection

Repository Link:

https://github.com/Mat6577/COS214_Practical5

Collaboration Reflection:

We began communicating as soon as the practical was released, which allowed us to divide the work equally. As a result, every member contributed to each component: writing code, designing the UML diagrams, and using a developer tool.

Despite a few minor merge conflicts and design changes along the way, we communicated well and successfully integrated our code into a final working product.

Team Contribution Statements

- Ashley: I implemented the Command and Mediator design patterns, set up the Docker build environment, worked on the initial UML and wrote the final submission PDF. Because my work involved offline architectural design and compiling the final report, my contributions are grouped into a few large commits rather than many small ones.
- Tebello: I worked on the main, Façade, and adapter, which explains the large gap between my commits and my teammates commits. The work was however, split equally, and each member implemented at least one pattern.
- Navela: I implemented the State design pattern to manage dynamic transitions between incident phases (Detection, Response, Recovery). Also designed the Facade sequence diagram and the final activity diagram for the UML documentation.